

Effect of feeding level on growth and nutrient deposition in rainbow trout (*Oncorhynchus mykiss* Walbaum) growing from 150 to 600 g

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Abstract

The effects of feeding level on growth and energy partitioning were studied in rainbow trout growing from 150 to 600 g. Triplicate groups of fish (initial weight 158 g fish⁻¹) were fed a practical diet at various feeding levels (25%, 50%, 75% and 100% of near satiation) for 24 weeks at 8.5 °C. The final body weights of fish were 235, 381, 526 and 621 g. Restricted feeding levels significantly reduced live weight gain. Feeding levels had less pronounced effects on feed efficiency ratio, which were 0.98, 1.08, 1.02 and 0.83, respectively, for the 25%, 50%, 75% and 100% feeding levels. The growth of fish fed to near satiation was accurately described by the thermal-unit growth coefficient. The growth data also showed that the widely used specific growth rate was not an appropriate model. Fish fed at the lowest feeding level (25%), which represented a maintenance ration (energy gain was less than 2 kJ fish⁻¹ day⁻¹), showed positive protein deposition but negative lipid deposition. This indicates that fish fed a maintenance ration mobilize body lipid reserve to support protein deposition. The efficiency of energy for growth (k_g) was estimated to be 0.63. The factorial multiple regression approach estimated that the partial efficiencies of metabolizable energy utilization for protein deposition (k_p) and lipid deposition (k_l) were 0.63 and 0.72, and that maintenance energy requirement was about 19 kJ (kg BW^{0.824})⁻¹, for rainbow trout held at 8.5 °C.

Keywords: rainbow trout, feed, growth, energy, protein, lipid

Introduction

The efficiency of feed utilization has a determinant effect on feed cost and waste output for animal production systems. A better understanding of the conversion of dietary inputs into fish biomass is, therefore, key to improving the economical and environmental sustainability of most fish culture operations.

There is considerable controversy regarding the effect of feeding level on growth and feed efficiency ratio. Growth in relation to feed intake has been described as either linear or curvilinear for various fish species (Xie, Cui, Yang & Liu 1997). The maximum feed efficiency ratio is reported to have been achieved at feeding levels below that required for maximum growth ($G_{\max}$) (Brett & Groves 1979; Storebakken & Austreng 1987b), or only attained at maximum feed intake ($R_{\max}$) and $G_{\max}$ (Talbot 1993). Conversely, there are studies showing that moderate restriction in feed allocation had no significant effects on feed efficiency ratio (Storebakken & Austreng 1987a; Azevedo, Cho & Bureau 1998; Rasmussen, Ostenfeld & McLean 2000; Van Ham, Berntssen, Imsland, Pappour, Wendelaar Bonga & Stefansson 2003). These studies suggest that feed efficiency ratio achieves its maximum at moderate feed restriction (as low as 50% of near satiation) and this optimum is maintained up to maximum voluntary feed intake of the fish. Differences in methodological approaches, feed composition, fish species, genetic potential, fish size and life stage and efficiency of nutrient or energy utilization may have contributed to the discrepancies in the results. Systematic and more rational investiga-

tion of the pattern and cost of nutrient deposition should be used to achieve a better understanding of the conversion of dietary input to fish growth.

Bioenergetics approaches have been effectively used to predict growth, feed ration, feed efficiency ratio and waste outputs of fish fed diets of varying nutritional composition and reared under different environments (Cho 1992; Cho & Bureau 1998; Kaushik 1998; Lupatsch & Kissil 1998; Cui & Xie 1999; Bureau, Kaushik & Cho 2002; Bureau, Gunther & Cho 2003; Zhou, Xie, Lei, Zhu & Yang 2005). These studies have generated valuable information that is yielding a progressively better understanding of growth and nutrient utilization in fish. However, many bioenergetics models do not take into account the different energetic efficiencies of protein and lipid depositions. There is a need to improve the current bioenergetics models through better description of efficiencies of nutrient deposition. A factorial multiple regression approach that simultaneously estimates efficiencies of protein and lipid deposition has been applied to carp (*Cyprinus carpio* L.) (Schwartz & Kirchgessner 1995), gilthead sea bream (*Sparus aurata*), European sea bass (*Dicentrarchus labrax*) and white grouper (*Epinephelus aeneus*) (Lupatsch, Kissil & Sklan 2003) and rainbow trout (*Oncorhynchus mykiss* Walbaum) (Rodehutscord & Pfeffer 1999; Azevedo, van Milgen, Leeson & Bureau 2005). Among these studies, the estimates for carp, gilthead sea bream, European sea bass and white grouper were obtained through studies with different feeding levels, whereas the studies conducted with rainbow trout were restricted to fish fed to satiation. It is necessary to examine the effect of feeding level on protein and lipid deposition in rainbow trout in order to improve the current bioenergetics models for rainbow trout. The objective of the present study was to examine the effect of feeding level on growth, feed efficiency ratio and patterns and cost of nutrient deposition in rainbow trout weighing between 150 and 600 g.

Materials and methods

Experimental diets

A practical diet (Table 1) was formulated to contain all nutrients in excess of levels recommended by NRC (1993). The ingredients were obtained from a local feed manufacturer (Martin Mills, Elmira, ON, Canada). The diets were mixed using a Hobart mixer (Hobart, Don Mills, ON, Canada) and pelleted using a laboratory steam pellet mill (California Pellet Mill,

Table 1 Formulation and composition of the experimental diet

	g kg ⁻¹
Ingredients	
Fish meal, herring, 68% CP	320
Corn gluten meal, 60% CP	240
Soybean meal, 48% CP	160
Whey	70
Celite	10
Vitamin premix*	10
Mineral premix†	10
Fish oil, herring	180
Total	1000
Analysed composition (dry basis)	
Dry matter (%)	96.0
Crude protein (%)	43.5
Lipid (%)	23.5
Ash (%)	9.1
Phosphorus (%)	1.1
Gross energy (kJ g ⁻¹)	24.2
Digestible protein (DP) (%)	38.3
Digestible energy (DE), kJ g ⁻¹	18.4
DP:DE (g MJ ⁻¹)	20.8

*Provides per kilogram of diet: retinyl acetate (vitamin A), 3000 IU; cholecalciferol (vitamin D), 2400 IU; all-rac- α -tocopheryl acetate (vitamin E), 60 IU; menadione sodium bisulphite (vitamin K), 1.2 mg; ascorbic acid polyphosphate (Stay CTM 25% ascorbic acid), 240 mg; cyanocobalamine (vitamin B12), 0.024 mg; D-biotin, 0.168 mg; choline chloride, 1200 mg; folic acid, 1.2 mg; niacin, 12 mg; D-calcium pantothenate, 26 mg; pyridoxine · HCl, 6 mg; riboflavin, 7.2 mg; thiamin · HCl, 1.2 mg.

†Provides per kilogram of diet: sodium chloride (NaCl, 39% Na, 61% Cl), 3077 mg; ferrous sulphate (FeSO₄ · 7H₂O, 20% Fe), 65 mg; manganese sulphate (MnSO₄ · 36% Mn), 89 mg; zinc sulphate (ZnSO₄ · 7H₂O, 40% Zn), 150 mg; copper sulphate (CuSO₄ · 5H₂O, 25% Cu), 28 mg; potassium iodide (KI, 24% K, 76% I), 11 mg.

San Francisco, CA, USA). The feed pellets were subsequently dried in a current of air at room temperature for 24 h, sieved and stored at 4 °C until use.

Fish, experimental conditions and feeding

Rainbow trout, average body weight (BW) 158 ± 1 g, were obtained from a domestic fall-spawning broodstock held at the Alma Aquaculture Research Station (Elora, ON, Canada). Fifty fish were randomly allocated to each of the 12 fibreglass tanks with a rearing volume of 90 L. The tanks were supplied with oxygenated and degassed groundwater at a constant year-round temperature of 8.5 °C. A photoperiod regime of 12 h light and 12 h was maintained by artificial lighting. The fish were treated in accordance with the guidelines of the Canadian Council on Animal Care

(CCAC 1984) and the University of Guelph Animal Care Committee.

The experimental diet was allocated to the fish according to four feeding levels (25%, 50%, 75% or 100%), with triplicate tanks per feeding level. The feed intake of the fish of treatment 100% determined the feed allocation of the fish of the other three treatments. The fish of treatment 100% were carefully hand fed to near satiety 2 days a week (Mondays and Thursdays), and fed with an automatic feeder in pre-allocated amounts the remaining 5 days of the week. The fish of other treatments were fed at feeding levels imposing various degrees of feed restriction (25%, 50% and 75%) relative to treatment 100%. The daily feed ration was delivered using vibrating automatic feeders (Sweeny Enterprises, Boerne, TX, USA), adjusted to deliver the ration in four approximately equal meals during the light period. Mortality and morbidity was checked daily. All fish were weighed every 4 weeks. Live weight gain, thermal-unit growth coefficient (TGC), feed efficiency ratio (live weight gain:feed dry weight) and percent mortality was calculated. After 12 weeks, the fish were transferred to 333 L tanks until the end of the experiment.

At the end of the growth trial, the digestibility of the experimental diet were determined at the Fish Nutrition Research Laboratory using the Guelph system (Cho, Slinger & Bayley 1982). Four faecal samples were collected from four different collection units comprising three tanks each. Each tank was stocked with 40 fish with an average BW of 600 g. The fish were acclimated to the experimental diet over 7 days, and faecal material were then collected according to Cho *et al.* (1982) for the next 5 days.

Sampling and chemical analyses

At the beginning of the trial, a pooled sample of 15 fish was collected to serve as an initial carcass sample. At the end of 24 weeks, four fish, respectively, were sampled at random from each tank and anaesthetized with t-amyl alcohol and killed with a cephalic blow. The fish were pooled, autoclaved, ground into homogeneous slurry, freeze-dried, reground and stored at -20°C until analysis. Diet, ingredients and carcass samples were analysed for dry matter (DM) and ash according to AOAC (1995), crude protein (CP, $\%N \times 6.25$) by the Kjeldahl method using a Kjeltech autoanalyser (Model 1030, Tecator, Höganäs, Sweden) and total lipid according to the method of Bligh and Dyer (1959). Gross energy (GE) content of

samples was measured using an automated bomb calorimeter (Model 1272, Parr Instruments, Moline, IL, USA). Phosphorus (P) content was analysed by a commercial laboratory (Labstat, Kitchener, ON, Canada) according to AOAC (1995).

Calculations

Growth rate was calculated using TGC according to the following equation (Iwama & Tautz 1981; Cho 1992):

$$\text{TGC} = \frac{[\text{FBW}^{(1/3)} - \text{IBW}^{(1/3)}]}{\left[\sum_{i=1}^d (T_i \times d) \right]} \times 100$$

where FBW is the final body weight (g), IBW the initial body weight (g), T_i the average temperature on the i th day ($^{\circ}\text{C}$) and d the number of days.

The apparent digestibility coefficients (ADC) for the nutrients and energy of the diet were calculated as follows (Cho *et al.* 1982):

$$\text{ADC} = 1 - (F/D \times D_i/F_i)$$

where D is the % nutrient (or kJ g^{-1} GE) of the diet; F the % nutrient (or kJ g^{-1} GE) of faeces; D_i the % digestion indicator (AIA) of the diet; and F_i % digestion indicator (AIA) of faeces.

Metabolizable energy (ME) was calculated as the difference between digestible energy (DE) and the energy as non-faecal N losses (urine energy UE+branchial energy ZE) (Cho & Kaushik 1990). Metabolic body weight was calculated as described by Azevedo *et al.* (1998).

The efficiency of energy utilization for growth (k_g) was calculated from the slope of the regression of recovered energy (RE) on ME intake. Maintenance energy requirement (MEM) was estimated by extrapolating the regression line to zero energy retention ($\text{RE} = 0$).

A factorial approach was further used to estimate simultaneously the MEM, the partial efficiency of protein deposition and lipid deposition (Kielanowski 1965):

$$\text{ME} = \text{MEM} + (1/k_p) \times \text{PD} + (1/k_l) \times \text{LD}$$

where ME is the metabolizable energy intake; MEM the maintenance energy requirement; k_p the partial efficiency of energy utilization for protein deposition; k_l the partial efficiency of energy utilization for lipid deposition; PD the energy retained as protein (kJ day^{-1}); and LD the energy retained as lipid (kJ day^{-1}), which was estimated as the energy

retained as non-protein, the difference between RE and energy retained as protein (PD).

Statistical analysis

Data were analysed as a complete random block design using the general linear model of the SAS/STAT software (SAS Institute, Cary, NC, USA). The means of dependent variables were compared using Tukey's honestly significant difference (HSD) test; significance was considered when $P < 0.05$. Dietary effects were examined using orthogonal contrast (Steel, Torrie & Dickey 1997). A factorial approach to estimate k_p , k_t and MEM was used by PROC NLIN using SAS/STAT software (SAS Institute).

Results

The apparent digestibility coefficients of the experimental diet were estimated to be 76% for DM, 88% for CP ($N \times 6.25$), 76% for GE, 60% for ash and 73% for P.

The growth curve of rainbow trout is depicted in Fig. 1. The TGC model faithfully described the live weight gain of fish fed to near satiation, whereas specific growth rate (SGR) resulted in overestimation of live weight of the fish when used to extrapolate growth data (Fig. 1b). Table 2 presents the growth performance and nutrient utilization of the fish. Restricted feedings resulted in a significantly lower live weight gain. Thermal-unit growth coefficient increased with the increment in feeding levels, but a significant quadratic effect was observed. Feed efficiency ratio was also affected by the treatment and responded in a curvilinear fashion. Feed efficiency ratio of fish fed at near satiation was significantly lower compared with restricted levels. Increased digestible N intake resulted in increased N retention; however, a significant quadratic effect suggested that the retention tended to level off at a high digestible N intake (Fig. 2). Restricted feeding levels linearly reduced retained lipid (g fish^{-1}) and the energy retained by fish (kJ fish^{-1}). Incremental feeding levels resulted in a linear increase in digestible P retained by fish (g fish^{-1}), but a linear decrease in P retention efficiency (% digestible P intake).

The carcass composition of fish is presented in Table 3. Fish fed the lower feeding levels had significantly higher moisture contents and lower GE contents. Restricted feeding levels reduced the concentration of lipid, and a quadratic effect was ob-

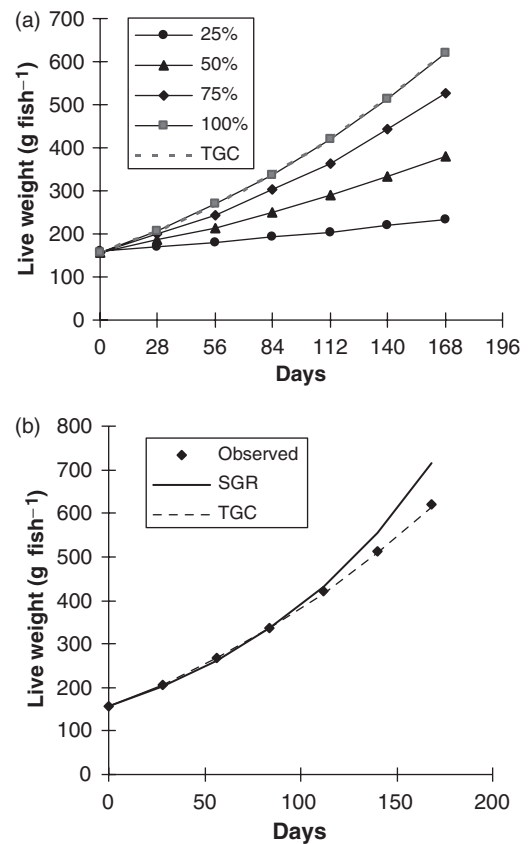


Figure 1 (a) Growth curve of rainbow trout fed at different feeding levels and reared at 8.5 °C for 24 weeks. (b) Simulation of growth of fish fed at satiation by thermal-unit growth coefficient and specific growth rate.

served. The protein content of the fish was not affected by feeding level. Restricted feeding levels resulted in a linear increase in ash contents of the fish. A significant linear increase in P concentration was observed with decreasing feeding levels.

Figure 3 presents the relationship between metabolizable energy intake and RE. The energy efficiency for growth (k_g) was $0.63 (\pm 0.03)$. By extrapolating to zero energy retention, the MEM was estimated to be $7.9 (\pm 1.7) \text{ kJ (kg BW}^{0.824})^{-1} \text{ day}^{-1}$. The factorial approach estimated ($P < 0.0001$, $R^2 = 0.98$) MEM to be $19.2 \pm 16.7 \text{ kJ (kg BW}^{0.824})^{-1}$, and the partial efficiency of protein deposition (k_p) and of lipid deposition (k_l) to be $0.63 (\pm 0.60)$ and $0.72 (\pm 0.25)$ respectively.

Discussion

The TGC model, which has been shown to describe very accurately the growth of salmonids of less than

Table 2 Growth, feed efficiency and utilization of nitrogen (N), lipid, phosphorus (P) and energy of rainbow trout* fed the experimental diets for 24 weeks

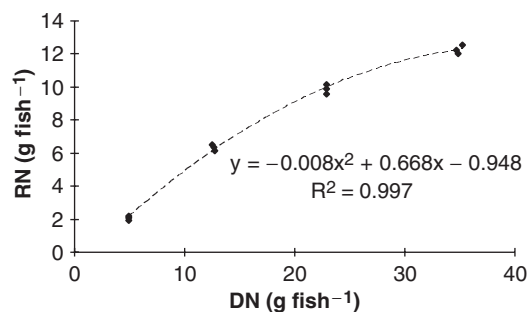
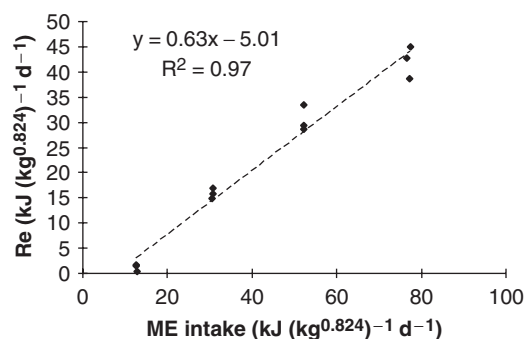
Parameters	Feeding level				Pooled SEM†	Contrast significance	
	25	50	75	100		Linear	Quadratic
Final body weight (g fish ⁻¹)	235	381	526	621	6	0.001	0.05
Gain, wet weight (g fish ⁻¹)	77	216	369	462	6	0.001	0.01
TGC	0.054	0.130	0.188	0.220	0.002	0.001	0.001
Feed, dry weight (g fish ⁻¹)	78	201	364	554	N/A‡	N/A‡	N/A‡
Feed efficiency ratio, gain:feed	0.98	1.08	1.02	0.83	0.01	0.001	0.001
Retained N (g fish ⁻¹)	2.1	6.3	9.9	12.3	0.1	0.001	0.001
N retention efficiency (% digestible N intake)	42	50	43	35	1	0.01	0.001
Retained lipid (g fish ⁻¹)	–4	20	53	73	1	0.001	NS
Retained energy (kJ fish ⁻¹)	126	1818	3691	5247	183	0.001	NS
Energy retention efficiency (% digestible energy intake)	9	50	56	52	3	0.001	0.001
Retained P (g fish ⁻¹)	0.41	0.96	1.43	1.69	0.05	0.001	0.05
P retention efficiency (% digestible P intake)	68	62	51	40	3	0.001	NS

*Initial body weight = 158 g fish⁻¹.

†Pooled standard error of a mean.

‡Not applicable, independent variable.

NS, non-significant; NA, not applicable, independent variable; TGC, thermal-unit growth coefficient.

**Figure 2** Nitrogen (N) utilization as a function of digestible N intake of rainbow trout reared at 8.5 °C for 24 weeks.**Figure 3** Recovered energy as a function of metabolizable energy intake of rainbow trout reared at 85 °C.**Table 3** Carcass composition of the rainbow trout* fed the experimental diet for 24 weeks

	25	50	75	100	SEM†	Linear	Quadratic
Moisture (%)	76.4	70.6	67.4	64.7	0.6	0.001	0.05
Crude protein (%)	16.9	17.5	16.8	16.7	0.2	NS	NS
Lipid (%)	4.2	8.7	12.6	13.9	0.5	0.001	0.05
Ash (%)	2.5	2.3	2.1	1.9	0.1	0.001	NS
Phosphorus (%)	0.45	0.42	0.39	0.37	0.02	0.01	NS
Gross energy (kJ g ⁻¹)	5.5	7.8	9.2	10.3	0.3	0.001	0.05

*Initial carcass composition: 72.3% moisture, 16.9% crude protein, 8.4% lipid, 2.3% ash, 0.40% phosphorus, 7.3 kJ g⁻¹ gross energy.

†Pooled standard error of a mean.

NS, non-significant.

100 g (Cho 1992), faithfully described the growth of rainbow trout of up to 600 g fed to near satiation in this study. Live weight of the fish increased curvilinearly, whereas the cubic root of live weight increased in a highly linear fashion, supporting the observations of Iwama and Tautz (1981). The growth of fish fed to near satiety also showed that the widely used instantaneous growth rate or SGR is not an appropriate growth model. It is widely recognized that SGR decreases with the size of the fish and the length of the time interval used in the calculation. The dependency of SGR on fish weight results in meaningless comparisons of growth rates among different

groups. The weakness of the SGR model is illustrated by Fig. 1b. When TGC and SGR of the first 84 days were extrapolated to 168 days of growth, the SGR model grossly overestimated the BW of the fish, whereas TGC provided an accurate prediction.

Feed intake had a significant effect on body composition in the present study. A reduction in feeding level produced fish with a lower lipid level and GE content. However, body protein content was not affected by the treatment. This is in agreement with Shearer (1994), who concluded that the protein content of growing salmonids is determined solely by fish size and is not affected by growth rate, diet or environmental factors. By comparison, lipid content is variable and is affected by endogenous and exogenous factors (Shearer 1994). In the present study, treatment had a significant effect on body lipid concentration; a drastic reduction in lipid level was observed at restricted feeding levels. Restricted feeding also resulted in a linear increase in body ash and P concentrations. This is probably because of the fact that feed intake could potentially result in changes in the proportion of bone mass and soft tissue mass. Fish fed with restricted rations may have less muscle mass and a greater bone mass; thus, percentage ash and P content should potentially be higher in fish fed at low feeding levels.

Nitrogen utilization was also found to be a quadratic function of digestible N intake. The digestible N was apparently more efficiently utilized at a low intake level than at a high N intake. The lowest feeding level (25%) represented approximately a maintenance ration (energy gain was less than $2 \text{ kJ fish}^{-1} \text{ day}^{-1}$). Fish fed at the maintenance ration still showed positive protein deposition but negative lipid deposition, as presented in Fig. 4. Similar observations have been reported for other farmed animals (Kyriazakis & Emmans 1992a, b; Boekholt, Vandergrinten, Schreurs, Los & Leffering 1994). This supports the hypothesis that live weight gain is driven by protein deposition and not by lipid deposition. In this study, fish fed on a maintenance ration still gained weight. Fish fed at the lowest feeding level (25%) still showed a feed efficiency ratio of about 1. The results also illustrate that the lipid:protein deposition ratio is not constant at different energy intake. A decreasing protein to lipid deposition ratio can be observed at feeding level approaching maximum protein deposition. As protein deposition is the main determinant of live weight gain of the animal, feed efficiency ratio should decrease as feeding level approaches and exceeds maximum protein deposi-

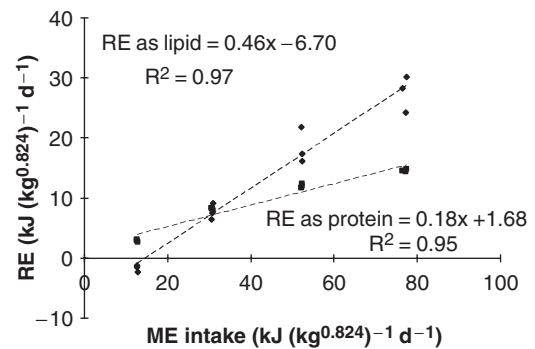


Figure 4 Recovered energy (RE) as protein and lipid (RE as lipid was calculated as the difference between RE and RE as protein) as a function of metabolizable energy (ME) intake of rainbow trout reared at 8.5 °C for 24 weeks.

tion potential (commonly referred to as PD_{max} in the animal nutrition literature). This has been confirmed by the significantly lower feed efficiency ratio of fish fed to near satiation (0.83) than the restricted feeding levels. The efficiency of DE intake for RE deposition, though, did not decline at satiation, indicating that the dietary energy was efficiently utilized at a high feeding level. This observation contradicts the assumption that disproportional higher metabolic costs are associated with higher feed consumption (Talbot 1993).

The observed effect of feeding level on feed efficiency ratio in this study demonstrates that the conversion of dietary input to fish biomass should be examined more systematically. Feed efficiency ratio is affected by many factors, such as feed composition, fish size and life stage, amino acid intakes and the efficiency of retention of these amino acids. One of the rather unique features of fish feeds, compared with terrestrial monogastric animal (e.g. poultry and swine) feeds, is that their nutritional composition varies considerably. The composition of fish feeds (i.e. CP, lipid, starch, energy levels) varies not only as a function of species and life stages for which they are formulated (trout vs. tilapia, starter vs. grower feed) but also as a function of a myriad of other factors (manufacturers' preference, production constraints, environmental constraints, consumer demands, economic climate, fish price, etc.). The amount of feed required by a fish to achieve a certain amount of weight gain depends primarily on the composition of the feed used. Mass of feed is not a measure of biologically available nutrients and energy. Consequently, the efficiency of feed utilization cannot be simply based on the mass of feed served. Biomass (live

weight) gain in animals is the result of deposition of water, protein, fat, minerals and a small amount of other components (glycogen, etc.). The amount of components deposited per unit of live weight gain is not constant but rather changes with weight of the animal, feed used, physiological state and other factors. The protein content of fish is found to remain relatively constant across life stage and hardly affected by dietary factors (provided the dietary essential amino acid is adequate); lipid content of fish is variable depending on the energy intake and growth (Shearer 1994). Consequently, the energetic efficiency of nutrient deposition also differs. As both 'feed' and 'fish biomass' are highly variable in terms of chemical composition, feed efficiency ratio can only be considered a semi-quantitative parameter. It should never be considered an absolute indicator of efficiency of conversion of dietary nutrients (inputs) into fish biomass. It is, therefore, hardly surprising to find contradictory evidences in the literature on the effect of feeding level on feed efficiency ratio. Great reserve must be used when comparing feed efficiency ratio across studies, farms, seasons and sampling intervals. Describing and predicting feed efficiency ratio achievable with different feeds and as a function of fish size, or environment is clearly not a simple task. A better understanding of the conversion of dietary input to fish growth can only come from a systematic and rational investigation of the pattern and cost of nutrient deposition.

As dietary nutrients are either incorporated as body components and/or utilized as 'fuel' for the various metabolic pathways involved in the deposition of the various body components, the study of the balance between dietary energy intake, expenditure and gain offers a relatively simple way of looking at dietary component utilization by animals (Bureau *et al.* 2002). Bioenergetics models are very useful and practical as they allow modelling of feed efficiency simply on the basis of DE content of the feed, and allow simple estimation of DE requirement of the fish. However, many bioenergetics models overlook the fact that protein and lipid depositions are achieved through different processes and have different effects on live weight gain of the animal. They also overlook the fact that protein and lipid depositions have different energetic efficiencies. It is necessary to improve the current bioenergetics models through estimates of energy partitioning of maintenance requirement and growth (protein and lipid deposition), and through estimates of energetic efficiencies of protein and lipid deposition.

In the present study, the utilization of ME intake was examined based on the regression of RE on ME intake. The efficiency of ME for production (k_g) in the present study was estimated to be 0.63. This is in agreement with most studies where k_g was estimated to be in the range of 0.40–0.70 (Meyer-Burgorff, Muller, Becker & Gunther 1989; Meyer-Burgorff, Osma & Gunther 1989; Cui & Liu 1990; Azevedo *et al.* 1998; Lupatsch, Kissil, Sklan & Pfeffer 1998; Ohta & Watanabe 1998; Rodehutsord & Pfeffer 1999; Lupatsch, Kissil & Sklan 2001, 2003). Recovered energy has been proven to be very useful as a measure of nutrient deposition and growth in most feed requirement models (Cho 1992; Cho & Bureau 1998; Cui & Xie 1999). However, RE is only an estimate of the deposition of energy-yielding nutrients (mostly protein and lipids) achieved through different systems and regulated by different factors. Consequently, RE measurements should always be accompanied by measurements of N gain (RN) and interpreted with care.

The efficiencies of ME for protein and lipid deposition were further examined through the factorial approach; k_p and k_f were estimated to be 0.63 and 0.72 respectively. The partial efficiencies were in line with estimates reported in the literature. Estimates of k_p were reported to range from 0.43 to 0.81 for different fish species, whereas estimates of k_f were between 0.60 and 0.90 (Schwartz & Kirchgessner 1995; Rodehutsord & Pfeffer 1999; Lupatsch *et al.* 2001, 2003; Azevedo *et al.* 2005).

Maintenance energy requirement was estimated to be $7.9 (\text{kg BW}^{0.824})^{-1}$ by regressing energy retention on ME intake and extrapolating ME intake for zero energy retention, whereas MEm estimated by the factorial approach was found to be 19.2 kJ $(\text{kg BW}^{0.824})^{-1}$. Cho and Kaushik (1990) suggested that MEm can, in practice, be estimated by measuring fasting heat loss (HEf). The empirical relationship has been established as equation $\text{HEf} = -1.04 + 3.26T - 0.05T^2$ kJ $(\text{kg BW}^{0.824})^{-1}$. This equation predicts that the HEf of rainbow trout reared at 8.5 °C should be around 23 kJ $(\text{kg BW}^{0.824})^{-1}$. This value is very close to the estimate of maintenance obtained with the factorial approach in this study. This suggests that the equation proposed by Cho and Kaushik (1990) is probably reliable enough to be used to estimate MEm of rainbow trout weighing up to 600 g. However, the large standard error associated with the estimates from the factorial approach warrants some caution. In this study, protein and lipid deposition is highly correlated ($r = 0.97$). The k_p and k_f estimates from this study should not be used blindly.

The factorial multiple regression approach could potentially yield nonsensical results because of the multicollinearity (high correlation among independent variables) (Slinker & Glianitz 1985; Birkett & de Lange 2001).

In summary, this study examined the effects of feeding level on growth and energy partitioning in large rainbow trout growing from 150 to 600 g. Restricted feeding levels significantly reduced live weight gain and growth rate, but the effect on feed efficiency ratio was less pronounced. Fish at the lowest feeding level (25%), which represented an approximate maintenance ration, showed positive protein deposition, negative lipid deposition and their feed efficiency ratio was close to 1. This indicates that fish were able to mobilize body lipid reserve to support protein deposition and gain weight. Feed efficiency ratio is a semi-quantitative parameter should not be considered an absolute indicator of efficiency of conversion of dietary nutrients inputs into fish biomass.

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